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North Eastern Mindanao State University

A DECISION SUPPORT SYSTEM ARCHIVING HUB:TRANSFORMING THESIS AND FEASIBILITY STUDY USING LLAMA

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CHAPTER I

INTRODUCTION

Background of the Study

The academic works found in University institutions at NEMSU Tandag Campus, such as thesis and feasibility studies, are vital intellectual resources. However, University institutions still use manual methods for archiving and retrieving these documents, which leads to inefficiencies, limited accessibility, and loss in documentation. This outdated process can make it challenging for students and faculty members to access valuable research materials easily for reference and further study. This study suggests creating a Decision Support System Archiving Hub using Llama to change the way academic works like thesis and feasibility studies are archived and accessed. The system aims to streamline document submission, storage, and retrieval processes by integrating modern digital archiving technologies to ensure fast, accurate, and secure handling of academic resources.

The Decision Support System Archiving Hub powered by Llama offers a centralized platform for systematically categorizing academic papers and making them readily available to users. Furthermore, the system will prioritize data privacy by implementing robust measures to protect sensitive academic information and

intellectual property. Access to the archived works will be carefully regulated, ensuring that only authorized users can retrieve documents. Encryption, user authentication, and secure storage protocols will be integrated to safeguard academic resources against unauthorized access, breaches, or data loss. Additionally, with the integration of Llama, the system will use data-driven insights to help identify gaps in existing academic work, allowing users to locate relevant resources and guiding institutions in managing research trends. This comprehensive approach not only improves accessibility but also supports better academic planning and resource management. The system's goal is to provide a comprehensive, efficient, and secure solution for academic document management, thereby enhancing the accessibility, organization, and protection of institutional research data.

Other studies express concerns regarding the transparency of the manual method of archiving theses and document studies. Researchers and students who seek specific materials in these archives often find traditional navigation methods to be tedious, time-consuming, and frustrating (Nitin Ghadge, 2024). A common issue in university-level research is the loss of data due to poor data management or the departure of essential staff members. Data is frequently stored on personal laptops, flash drives, or other devices that are not part of a centralized system, increasing the risk of mismanagement or loss (Wright et al., 2020). Moreover,

confusion surrounding the elements of research papers remains prevalent among students. The key to writing an effective research paper lies in understanding these common elements and their definitions, though many students still find it challenging (Alkhaqani, 2022).

Another issue is the reliance on traditional teaching materials, which no longer engage students. Despite the availability of online journals and websites, students often prefer unverified articles because they appear quickly in search engine results, neglecting more reliable sources (Febriati et al., 2020). Combining these challenges with a Decision Support System (DSS) Archiving Hub utilizing Llama's advanced capabilities can transform how academic works such as theses and feasibility studies are archived and accessed, providing an efficient and centralized solution.

Academic institutions have implemented digital solutions for storing documents, but challenges persist in efficiently managing large volumes of research papers such as theses and feasibility studies. Manual paper-based archiving creates inefficiencies, leading to cumbersome access and retrieval. Previous studies on digital archiving systems often overlook the need for intelligent data organization and decision support tools required to enhance accessibility to academic research.To address these shortcomings, this study presents the Decision Support System Archiving Hub enhanced by Llama, a cutting-edge

platform aimed at revolutionizing the management and retrieval of academic works. By leveraging Llama's advanced capabilities, the system will assist in identifying gaps in existing academic work, helping users locate relevant resources while guiding institutions in managing research trends. This platform offers a user-friendly experience, allowing students and faculty to access research materials efficiently through a web-based interface. The Decision Support System Archiving Hub ensures rapid retrieval, secure storage, and organized archiving, ultimately transforming the accessibility and usability of academic archives.

In addressing the inefficiencies of traditional academic archiving systems at NEMSU Tandag Campus, this study proposes the Decision Support System Archiving Hub, a decision support system that redefines how theses and feasibility studies are stored and accessed. The system introduces features like automated archiving, intelligent categorization, and data-driven insights powered by Llama's tools to identify gaps in existing research, helping users locate relevant resources efficiently. This platform enables students and faculty to submit and retrieve research papers through an intuitive web interface. Unlike conventional systems, the Decision Support System Archiving Hub integrates decision support tools to provide real-time analytics on academic trends, enhancing both resource management and institutional planning. This innovative solution is tailored to the unique needs of academic research at NEMSU Tandag Campus, transforming

storage and accessibility while fostering a more organized and efficient research environment.

Theoretical Framework

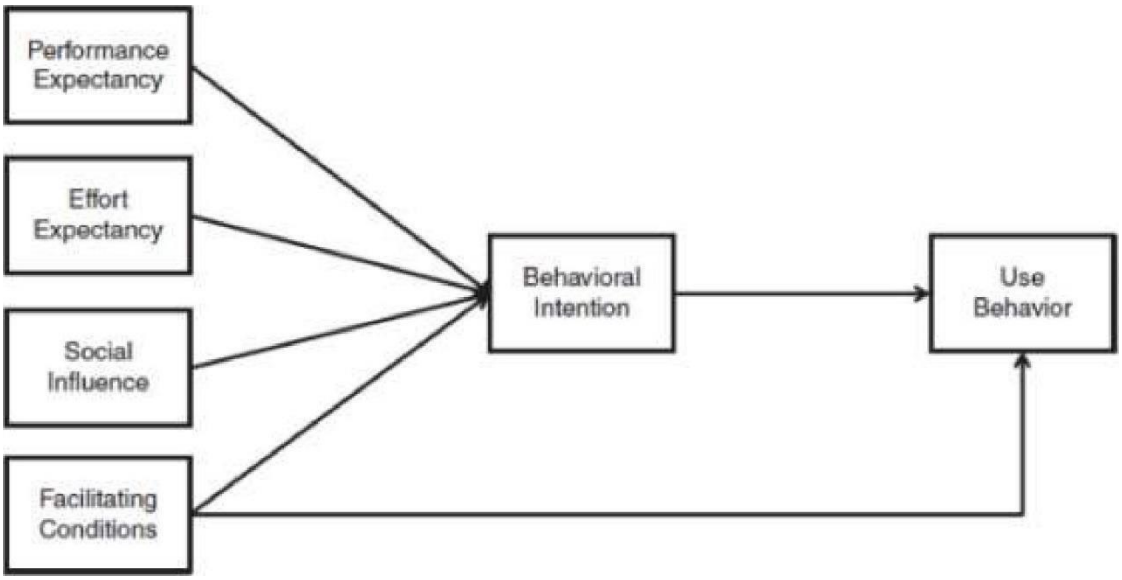

Figure 1. The Unified Theory of Acceptance and Use of Technology (UTAUT) Model (Suki* & Suki, 2018).

The UTAUT model, according to (Suki* & Suki, 2018) is a foundational framework for understanding the factors that influence the acceptance and use of the Decision Support System Archiving Hub, encompassing four primary constructs: Performance Expectancy, Effort Expectancy, Social Influence, and

Facilitating Conditions. Performance Expectancy represents users' perceptions of the system's effectiveness in enhancing their academic research and management, while Effort Expectancy relates to the ease of navigating the system, significantly impacting users' willingness to adopt it.

In the context of NEMSU Tandag Campus, the UTAUT model facilitates a structured exploration of how students and faculty perceive the benefits of the Decision Support System Archiving Hub. By identifying the factors that shape attitudes and intentions toward the archiving system, the model provides valuable insights for tailoring features and support to promote user engagement.

Through the application of the UTAUT model, this study sets the stage for introducing the Decision Support System Archiving Hub, an innovative platform designed to meet the unique needs of academic institutions. Grounded in the principles of U TAUT, this system aims to be user-friendly, enhancing the accessibility and organization of academic research at NEMSU Tandag Campus.

Figure 2. Theoretical Framework UTAUT model Support.

The adoption of the DSS Archiving Hub can be better understood through the UTAUT framework, which focuses on the key factors that influence whether people will use a system. Performance expectancy is an important factor, as users believe the system will help them work faster and more accurately when archiving and retrieving documents. The system also reduces mistakes and saves time, making work more efficient. Another key factor is effort expectancy, which looks at how easy the system is to use. With a simple design and easy navigation, users are more likely to feel confident and satisfied using the system.

Social influence also plays a role in adopting the system. When colleagues, supervisors, and leaders encourage the use of the system, it motivates users to embrace it. Following workplace norms also reinforces its importance. Facilitating conditions, such as having the right resources, infrastructure, and technical support, make it easier for users to adopt the system. Integration with existing technologies and training further smooth the transition.

When all these factors come together, users decide to adopt the system, using it regularly for their tasks. Over time, this leads to positive feedback and greater engagement. Ultimately, the adoption of the system results in a measurable impact on efficiency, with improved productivity and smoother workflows. Users

spend less time archiving and retrieving documents, make fewer mistakes, and experience better overall performance, demonstrating the system's effectiveness in meeting organizational needs.

Conceptual Framework

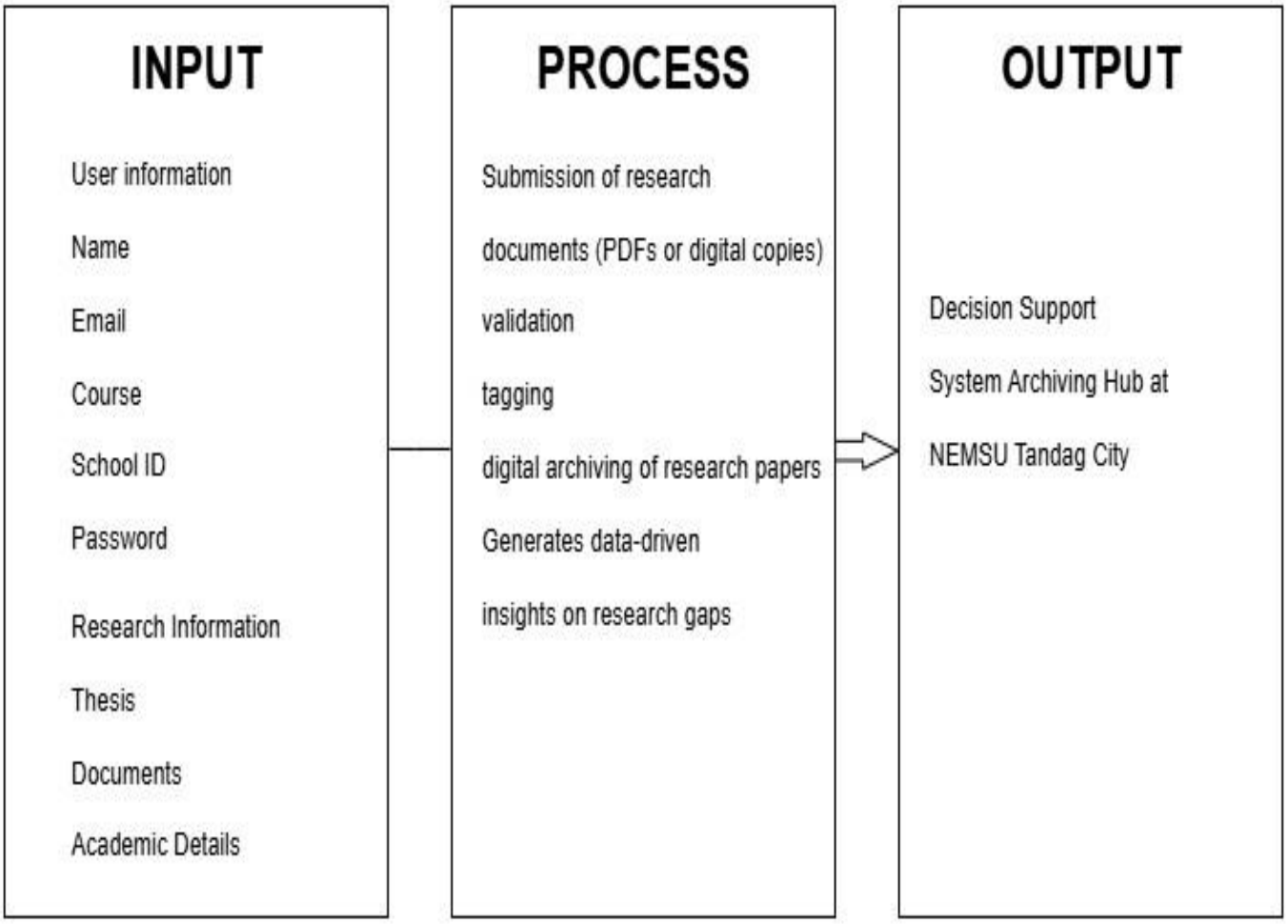

Figure 3. Conceptual Framework of the Study

The conceptual framework for this research illustrates the processing of user and document input to create an intelligent and accessible repository for academic works.

The process begins with the input phase, where essential user and research data are gathered. This includes personal information such as name, email, course, school ID, and password, as well as research-specific details like thesis title, keywords, abstract, and supporting documents.

During the process phase, the system validates the user's NEMSU account and securely stores the submitted information. The system integrates Llama to analyze the thesis abstract and identify research gaps. This AI-powered process enables intelligent tagging and categorization of academic works, ensuring that each document is systematically archived and organized for easy retrieval.

In addition, the system monitors and maintains older research documents to ensure continued accessibility and relevance. Thesis works older than five years are reviewed and managed to prevent loss or degradation of valuable research materials.

In the output phase, the system delivers a functional Decision Support System Archiving Hub. This platform provides a centralized, web-based solution for archiving, searching, and retrieving theses and feasibility studies. With its

userfriendly interface and AI-powered capabilities, the system streamlines the management of academic documents, enhances search accuracy, and improves access to research materials for students and faculty at NEMSU Tandag City.

The framework demonstrates how user data, AI-driven analysis, and systematic document management interact to create a robust and efficient solution for storing and retrieving academic research.

Statement of the Problem

Within numerous academic institutions, the management of Thesis, and Feasibility Study document is a critical issue. This study seeks to address these issues by investigating potential improvements to the present archiving procedure at NEMSU Tandag Campus. Specifically, the study intended to answer the following questions:

1. How is the archiving process for thesis and feasibility study documents currently managed?

2. What processes can be implemented to validate, tag, and digitally archive research documents?

3. How can the system provide centralized, secure, and organized access to archived research works?

Objective of the Study

The fundamental goal of this research is to design, implement, and evaluate a website-based system equipped with AI Recommender to streamline the archiving process for thesis documents. To attain this general purpose, the study pursued the specific objectives listed below:

1. Analyze NEMSU Tandag Campus Thesis archiving process, including the author eligibility, publishing procedures, and the retrieval process.

2. To implement processes for validating, tagging, and securely archiving research documents.

3. To create a centralized and accessible platform for managing academic archives.

Scope and Limitations

The research project, titled A DECISION SUPPORT SYSTEM Archiving Hub: Transforming Theses & Feasibility Study Using Llama, aims to improve the retrieval and storage of academic theses at North Eastern Mindanao State University (NEMSU) Main Campus. The project focuses solely on analyzing thesis abstracts to identify research gaps using artificial intelligence, specifically Llama.

The project scope includes the implementation of Llama to process and analyze the abstract of each thesis to determine gaps in research and trends based on the submitted content. By examining only the thesis abstracts, the system provides valuable insights into existing studies, guiding future research directions.

The system will not generate, process, or display the full content of theses or feasibility studies; it will focus exclusively on the abstract to identify research gaps.

However, the study faces certain limitations. The system relies on abstracts alone to identify research gaps, which means that any missing or unclear information in the abstract may affect the accuracy of the analysis. Additionally, the system's performance may be limited if the training data lacks representation across all academic disciplines. Variations in user competence, including differences in digital literacy and familiarity with the platform, may also impact the system's usability.

Furthermore, only PDF file formats will be accepted for thesis abstract submissions to ensure standardized document processing and analysis. Despite these limitations, the project aims to deliver an efficient and AI-powered solution for analyzing thesis abstracts and identifying research gaps, ultimately enhancing academic resource management at NEMSU.

Significance of the Study

This study carries significant implications for various stakeholders in the academic research archiving process:

Administrators: Responsible for overseeing the system's operations, ensuring efficient and secure management of academic documents. They handle data organization, storage, and retrieval while monitoring system performance.

Students and Researchers: Gain improved access to theses and feasibility studies, allowing them to identify research gaps, retrieve relevant works faster, and streamline their academic efforts.

Librarians: Play a vital role in organizing, archiving, and preserving academic documents within the system. The system enables them to efficiently manage and maintain research materials, ensuring accessibility and proper handling of resources.

Faculty Members: Benefit from the system's data-driven insights for decision making, enabling better research trend analysis and resource allocation.

Academic Institutions: The Decision Support System Archiving Hub enhances document management practices, ensuring reliable, secure, and organized storage of research materials. This promotes a research conducive environment.

Future Researchers: This study serves as a reference for future projects focusing on improving digital archiving, helping researchers refine academic resource management.

Definition of Terms

Archiving Hub: A centralized platform for storing, managing, and retrieving academic research documents, such as theses and feasibility studies. This system allows users to access research abstracts and identify research trends efficiently.

Categorization: The process of organizing documents into specific, defined categories for easier access and retrieval.

Data-driven Insights: Analytical information derived from data, used to enhance decision-making and system functionality.

Data Processing: The process of collecting, organizing, and analyzing data to generate useful insights. In this study, it involves analyzing thesis abstracts to identify research gaps and categorize trends.

DSS (Decision Support System): A computer-based system that helps in decision-making processes by analyzing data and generating actionable insights.

Effort Expectancy: The perceived ease of using the system, which affects how willingly users adopt the platform.

Facilitating Conditions: The availability of resources and support systems that make it easier for users to adopt and use the technology effectively.

Feasibility Study: A detailed analysis that examines the viability of a proposed project or plan.

Llama: An AI-powered large language model (LLM) designed to process and analyze textual data. In this study, Llama is used to process thesis abstracts, identify gaps in research, and suggest trends for academic improvement.

NEMSU (North Eastern Mindanao State University): The institution where the study and system implementation are being conducted.

Performance Expectancy: Users' belief that the system will improve their academic productivity and research capabilities.

Research Gap: Refers to unexplored or underdeveloped areas within a particular field of study. The system identifies research gaps by analyzing the content of thesis abstracts, highlighting areas that need further investigation.

Research Trends: Patterns or directions emerging from a body of research. The system identifies these trends by analyzing multiple thesis abstracts to support future research planning.

Social Influence: The degree to which peers, mentors, or other influential individuals affect users' decisions to adopt the system.

Thesis: A document submitted by students to complete their degree, presenting original research on a particular subject.

Thesis Abstract: A brief summary of a research paper, typically including the research problem, objectives, methodology, and findings. The system uses this portion to identify gaps and trends within submitted theses.

Unified Theory of Acceptance and Use of Technology (UTAUT): A theoretical framework that explains user intentions to adopt a technology and their subsequent usage behavior. UTAUT identifies four key constructs: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions.

User Interface (UI): The visual elements and layout of a system that allow users to interact with it.

Web-based Platform: A software application accessed and operated through a web browser, allowing users to interact with the system online.

CHAPTER II

Review of Related Literature

In order to have a better idea of the topic and a good start in making the study, the researchers conducted the intensive readings both foreign and local from different resources. These sources include articles and case studies from the internet, and other reading materials. The review of related literature shall serve as a foundation to fully understand and meaningfully interpret the results of the ongoing study.

Foreign

The difficulties students encounter when choosing suitable thesis subjects for their courses of study. The authors point out that students may become confused and find it challenging to select appropriate and pertinent themes if the traditional manual method of choosing thesis fields is still used. In order to solve this problem, they created a decision support system that gives students accurate and mathematically supported suggestions for their thesis fields using a profile matching technique. The black box testing approach was used to thoroughly examine this system in order to guarantee its efficacy and dependability. The study's conclusions highlight how decision support tools can help students make

better decisions, which will benefit their overall academic experience when developing their theses (Ulsa, A., Sari, M., & Febriyanto, A.,2023).

In order to improve the management of academic publications, particularly theses and graduation projects, their study aimed on building a web-based library system. The system offers features for administrators and students and was built using ASP.NET and a SQL database. Administrators control institutional data, including departments and colleges, while students can browse documents with guest access or contribute research with member access. This system reduces human labor in university libraries and fulfills a requirement for effective, easily accessible academic archiving systems by streamlining document access and storage (Ibrahim, H., & Jadaan, F.,2020).

It highlights the practical benefits of archiving hubs, such as enhancing accessibility, organization, and long-term availability of academic data. Krefft underscores the necessity of structured data management, especially in academic settings where data from research and projects need to be preserved for future use. By centralizing data resources, Decision Support System solutions can improve research workflow and collaboration. The study shows how wellorganized data archives foster more productive academic communities by reducing the effort

required to access previous work. This centralization of data also enables efficient retrieval and minimizes redundancy. As a comprehensive solution, Decision Support System serves both students and faculty, helping them locate needed resources quickly. It concludes that digital archiving is a fundamental requirement for academic institutions aiming to optimize data management and research efficacy (Krefft, J.,2021).

The efficient retrieval and utilization of information from meeting recordings and transcripts are crucial in many fields, such as business, academia, and research meetings are important for cooperation, decisionmaking, and information sharing. The recordings and transcripts of meetings are significant resources for future reference and analysis nevertheless, the traditional techniques utilized for accessing and extracting information from meeting recordings can require a significant amount of effort and time. When the user needs to look back on particular information from the meeting recording, they cannot recall where exactly it is located or what exact keyword they have to use to search for it, resulting in difficulties when trying to navigate through extensive amounts of content to find specific insights or debates (Nitin Ghadge, 2024).

As research becomes more and more data intensive, managing this data becomes a major challenge in any organization. At university level there is seldom a unified data management system in place. The general approach to storing data in such environments is to deploy network storage. Each member can store their data organized to their own likings in their dedicated location on the network. Additionally, users tend to store data in distributed manner such as on private devices, portable storage, or public and private repositories. Adding to this complexity, it is common for university departments to have high fluctuation of staff, resulting in major loss of information and data on an employee's departure. A common scenario then is that it is known that certain data has already been created via experiments or simulation. However, it cannot be retrieved, resulting in a repetition of generation, which is costly and timeconsuming. Additionally, as of recent years, publishers and funding agencies insist on storing, sharing, and reusing existing research data. We show how digital preservation can help group leaders and their employees cope with these issues, by introducing our own archival system OntoRAIS(Wright et al., 2020).

Local

State that the study assessed the e-library's usability, dependability, performance efficiency, and functional compatibility, among other significant aspects. By providing theses and capstone projects as a valuable resource, the elibrary effectively satisfied the research needs of its users, according to the findings. This encouraging response demonstrates the importance of usercentered design in academic systems and implies that technologies can greatly improve the research experience when they are customized to specific user needs. Furthermore, the study underscored the significance of continuous assessments to guarantee that the e-library keeps adapting to user needs, which is consistent with the goals of the suggested decision assistance system for choosing thesis subjects in this study (Dela Cruz, J., Albino, R., Moraäa, L., Bachillar, E., & Dullas Jr., R.,2021).

The NIPSC Library's manual thesis and dissertation management inefficiencies and how a contemporary electronic archiving system may be implemented. Librarians and researchers may access academic materials across wired and wireless networks with ease because of the system's N-tier architecture, which also allows users to read related literature in real time. The addition of SMS functionality improves the user experience even more by giving users instant alerts

about material availability. Alpasan's results highlight that the system that was designed guarantees adherence to intellectual property and copyright laws while also increasing the effectiveness of resource management. The importance of technology in changing library services is emphasized by this study, which also emphasizes how academic institutions must embrace digital solutions to improve resource accessibility (Alpasan, F.,2022).

This study explores how online database resources are being used in a private Philippine university, focusing on accessibility, usability, and the impact on research and learning. The findings suggest that access to online databases increases student engagement and supports academic research by offering vast resources beyond traditional libraries. It underscores the importance of digital resources for educational institutions in the Philippines, where limited physical libraries can constrain research. Online databases empower students to perform research from anywhere, fostering a more independent approach to learning. The authors emphasize that digital archives not only enhance research capacity but also promote inclusivity by providing resources to those with limited access to physical libraries. The study concludes that a dedicated digital archiving system would further support research productivity, allowing easier access to past theses and studies. It also mentions that institutions should consider investing in user

training to maximize the benefits of these resources. The study's insights are relevant to digital archiving initiatives aimed at improving academic accessibility(Asio, J. M. R. , & Jimenez, E. C.,2020).

The study reveals that digital libraries are particularly beneficial in supplementing classroom materials, providing diverse resources that traditional libraries may lack. Alipio identifies challenges, including limited digital access and resource management skills, that can hinder the effective use of online libraries. However, the study finds that students and faculty who utilize these resources report improved research efficiency and greater exposure to global academic work. The author argues that a more robust, institutionalized digital archiving hub could further strengthen academic efforts, especially in environments where physical resources are limited. Digital libraries also support collaborative learning, as they allow for easy sharing of materials among students and educators. The paper advocates for more support from institutions to maximize digital libraries' benefits, including infrastructure improvements and training. These findings underscore the potential of a Decision Support System archiving hub to enhance access to critical academic resources across Philippine universities (Alipio, M. M., 2021).

The study outlines the structural and technical requirements necessary to implement effective digital archiving, highlighting the need for customized solutions tailored to institutional goals. Gonzales emphasizes that digital archiving systems simplify the storage and retrieval of academic works, particularly for theses, which are essential but often underutilized resources. The research identifies various challenges, such as funding constraints and technical expertise, which can impact the system's sustainability. Through case studies, Gonzales shows how a structured digital archiving approach improves resource accessibility for students and faculty. The study concludes that digital archiving systems are critical for preserving academic work in Philippine universities, enabling wider resource availability. Gonzales recommends further investment in training and system maintenance to maximize the effectiveness of these digital archiving solutions. This study is highly relevant to the goals of a Decision Support System archiving hub for thesis preservation and access(Gonzales, P. F., 2019).

Synthesis

The literature reviewed, it became clear that digital archiving and decision support systems (DSS) play a crucial role in enhancing the accessibility, organization, and usability of academic resources. These systems help address common issues such as inefficient data management, difficulty in thesis topic selection, and the challenges of manual resource handling in academic institutions. The studies highlight how Decision Support System can streamline research processes, reduce redundancy, and support students and faculty in accessing valuable resources more efficiently. Moreover, the adoption of centralized digital archiving systems fosters a more productive research environment, enabling easier access to previous work and contributing to a more collaborative academic community. These insights underscore the potential of a Decision Support System-based archiving hub to transform thesis and feasibility study management, supporting both academic productivity and long-term resource preservation.

CHAPTER III

Methodology

This chapter outlines the methodology employed to develop the Archiving Hub. It details the research design, system development framework, tools and technologies, data collection techniques, and evaluation methods. This concise overview ensures the reliability of the study and provides a comprehensive understanding of the development and implementation process of the Archiving Hub.

Research Design

Methodical Planning, Risk Analysis, Engineering and Evaluation are necessary for the creation of development NEMSU Archiving Hub. The Spiral Model, similar to the incremental model, places greater emphasis on risk analysis during the system analysis phase. It allows the system to progress through these stages in multiple cycles, ensuring continuous refinement and improvement (Kyeremeh, 2021). Because of its adaptability and iterative nature, which allow the system to change and grow with each iteration, the Spiral Model was selected for this project.

Figure 4. General Overview of Spiral Model of the Study(Sumaiya Simran, 2024).

The Spiral Model phases according to the Diagram above are:

Phase I – Planning (Requirement Gathering): In the planning phase, the requirements for the NEMSU Archiving Hub are gathered from stakeholders, including students, faculty, and library staff. This involves pinpointing particular difficulties, like the need for effective gap analysis and the difficulty in obtaining resources. To guarantee that the system meets user requirements and institutional objectives, the project's scope, objectives, resources, and beginning timetable are established during this phase.

Phase II – Risk Analysis (Risk Management): The risk analysis phase involves figuring out potential risks that might have an impact on the creation and operation of the NEMSU Archiving Hub. These risks could include issues with data privacy, user reluctance to embrace the new system, or technical constraints. To solve these problems, mitigation techniques are developed, including putting strong data security measures in place, holding training sessions, and guaranteeing scalability.

Phase III – Engineering (Designing and Coding Phase): In the engineering phase, the actual development of the NEMSU Archiving Hub takes place. This includes designing the system architecture, developing core features such as advanced search capabilities and gap identification, and integrating a secure database for archiving.

System Flowchart

This section displays the system flow for the NEMSU Archiving Hub. Both authors and students:

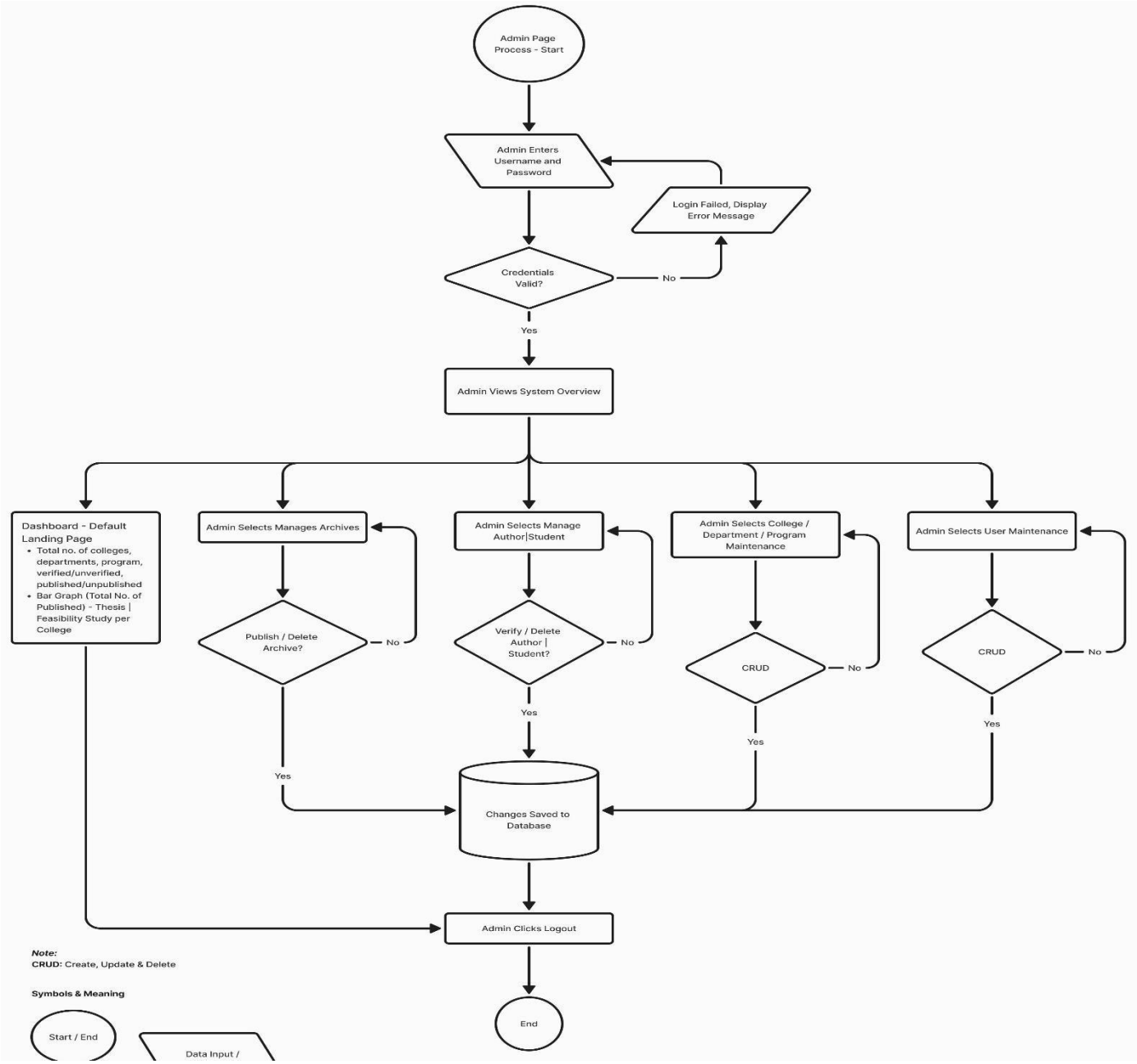

Figure 5. Archiving Hub Admin Process Flowchart

The process begins with the admin accessing the login page and entering their credentials. If the credentials are invalid, the system displays an error message; otherwise, the admin is granted access to the System Overview. From there, the admin can navigate to different sections, such as managing archives, verifying or deleting author and student records, and performing CRUD (Create, Read, Update, Delete) operations for colleges, departments, programs, and user maintenance. Changes made within these sections are saved directly to the database. Once tasks are completed, the admin can securely log out, ending the session. This flowchart highlights a clear and organized workflow for effective system administration.

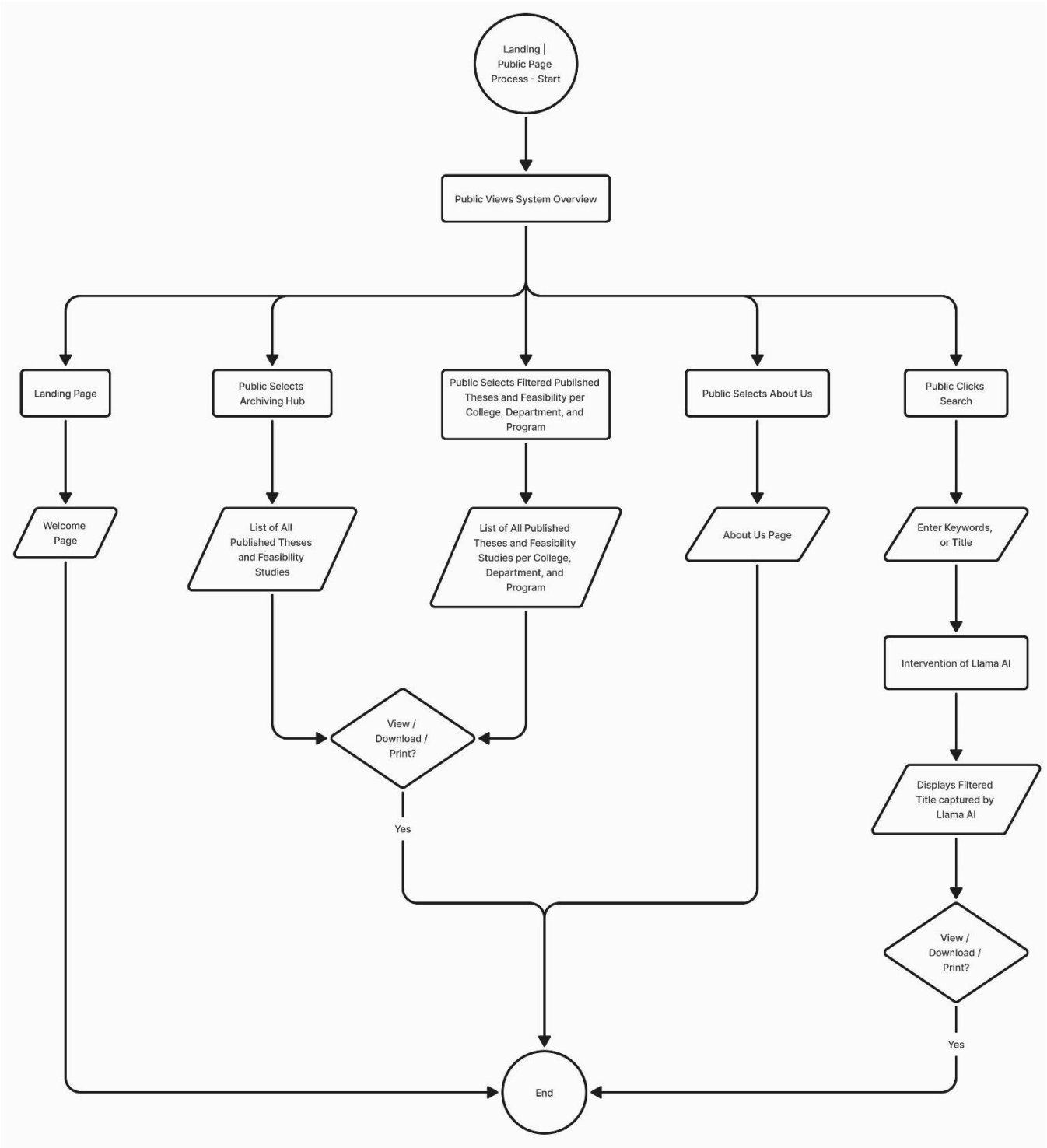

Figure 6. Archiving Hub Landing Page Process Flowchart

The process begins at the Landing Page, where the public views the System Overview. Users can navigate to several options: they may visit the Landing Page to access the Welcome Page, or select the Archiving Hub to view a list of all published theses and feasibility studies. Alternatively, users can filter Published Theses and Feasibility Studies by college, department, and program. They are then given the option to Download or Print the document.

Users may also select the About Us section to learn about the system or use the Search Feature by entering keywords or titles. The system, through Llama AI, processes the search input and displays the filtered results, allowing users to View, Download, or Print the retrieved data. Finally, all processes conclude when the user completes their actions, leading to the End of the session.

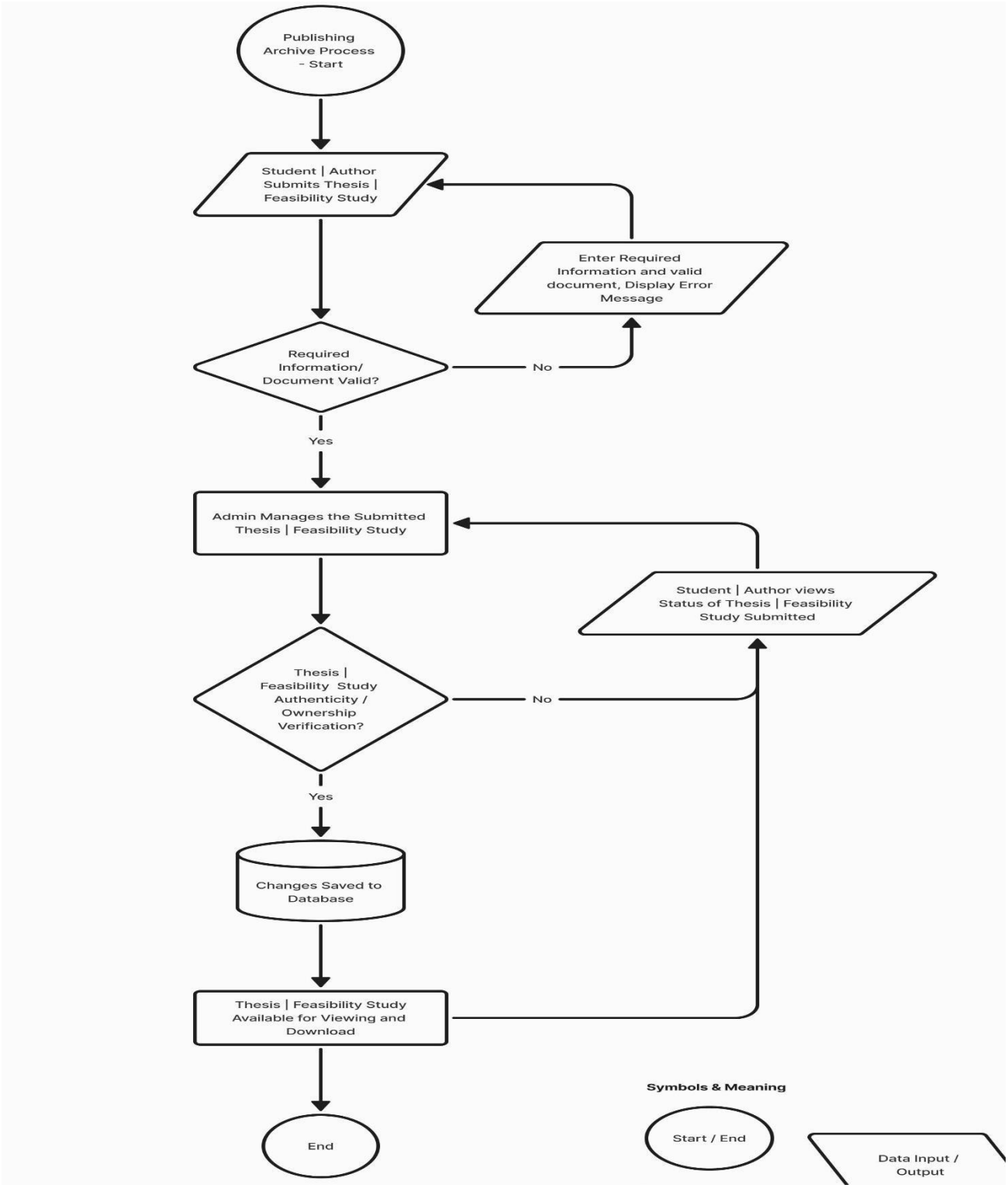

Figure 7. Publishing Archive Process Flowchart

The process begins when a Student or Author submits a Thesis or Feasibility Study for publication. The system checks if the Required Information and Document are valid. If invalid, the system prompts the user to Enter the Required Information and Valid Document, displaying an error message. Once validated, the Admin Manages the Submitted Thesis or Feasibility Study.

The admin then performs Authenticity and Ownership Verification of the document. If the document fails verification, the student or author can View the Status of their submission. If successful, the Changes are Saved to the Database, and the Thesis or Feasibility Study Becomes Available for Viewing and Download. The process ends upon successful publication of the document.

Algorithm Design

The algorithm design is crucial for the efficient management, storage, and retrieval of academic theses and feasibility studies within **the** Nemsu Archiving Hub. This section outlines the design of the algorithm used to manage and process academic documents, ensuring data integrity, organization, and ease of access.

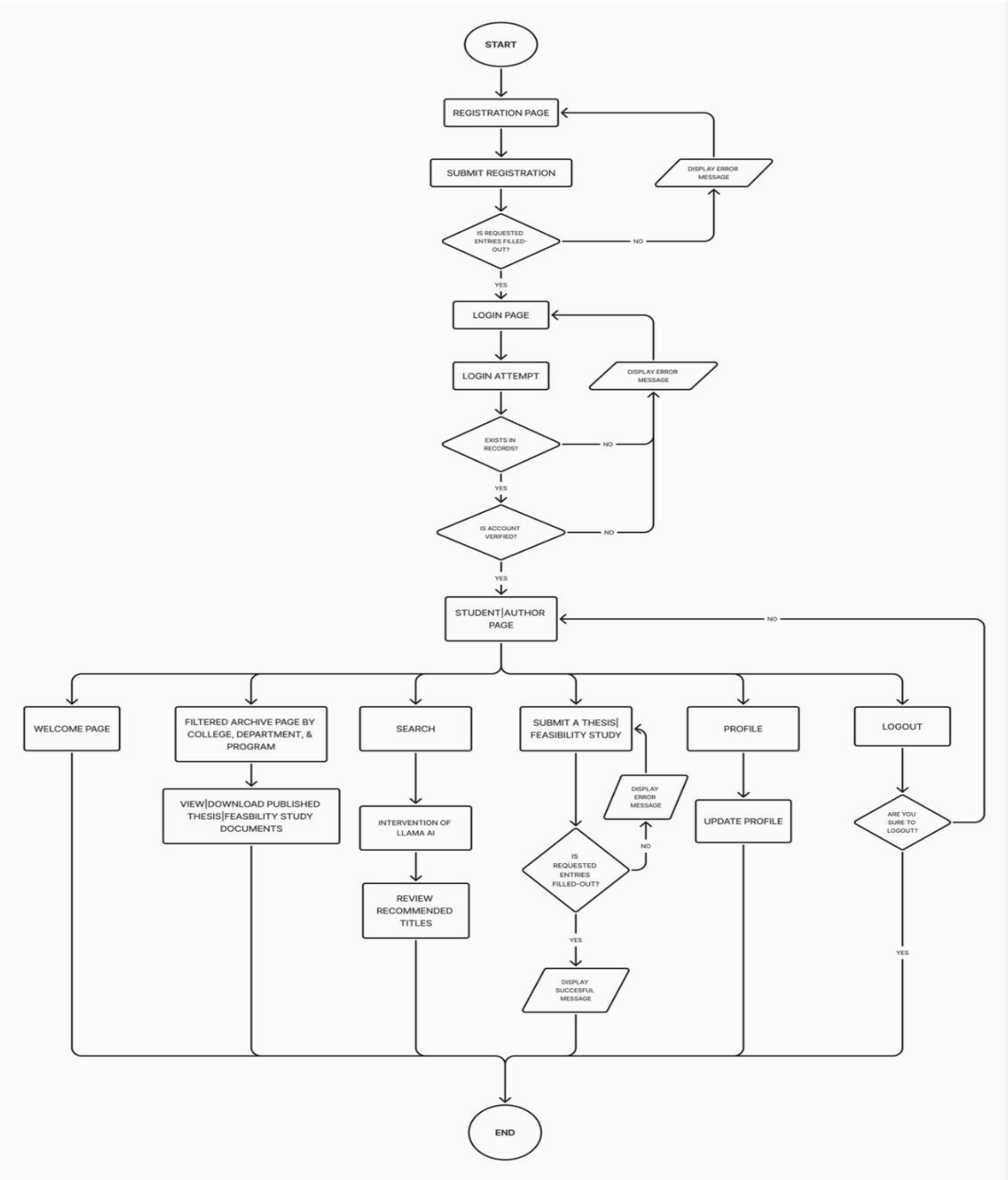

Figure 8. Archiving Hub Student and Author Page Process Flowchart

The figure illustrates the Student/Author Flowchart, where users interact with the system through a series of processes. The process begins by asking users whether they have an account. If they do not, they are directed to the Registration Page, where they must provide the required information. Once the registration form is submitted, the system verifies whether all mandatory fields have been filled. If the validation is successful, the user is redirected to the Login Page; otherwise, an error message appears, prompting the user to complete any missing fields. For registered users, the system requires login credentials, which are validated against the database to determine their correctness and account verification status. Users are granted access to the Student/Author Page upon successful login; otherwise, an error notice informs them of invalid credentials.

Once on the Student/Author Page, users have access to multiple features: The Home Page serves as the default landing page and provides general information about the system. Users can navigate to the Archives Page, where they can view, download, or print published theses and feasibility studies. For better organization, the system allows filtering based on college, department, or program. Additionally, users can access the Search Page, where they enter keywords or categories to retrieve specific studies. The system employs an intelligent filtering mechanism to display relevant results for the user to view or download. Users also have the option to submit their work through the Submit Page. By filling in all required fields, users can upload

theses or feasibility studies. If all inputs are valid, a success message is displayed; otherwise, an error prompts the user to correct missing or invalid information.

The system ensures personalization and accuracy through the Profile Page, where users can manage submitted archives, update their profile, or make necessary changes, which are saved to the database. Finally, users can choose to exit the system using the Logout Option, which asks for confirmation before ending the session. This well-structured flow ensures a seamless and user-friendly experience, meeting the needs of both students and authors by providing easy navigation, efficient submission, and quick access to academic resources.

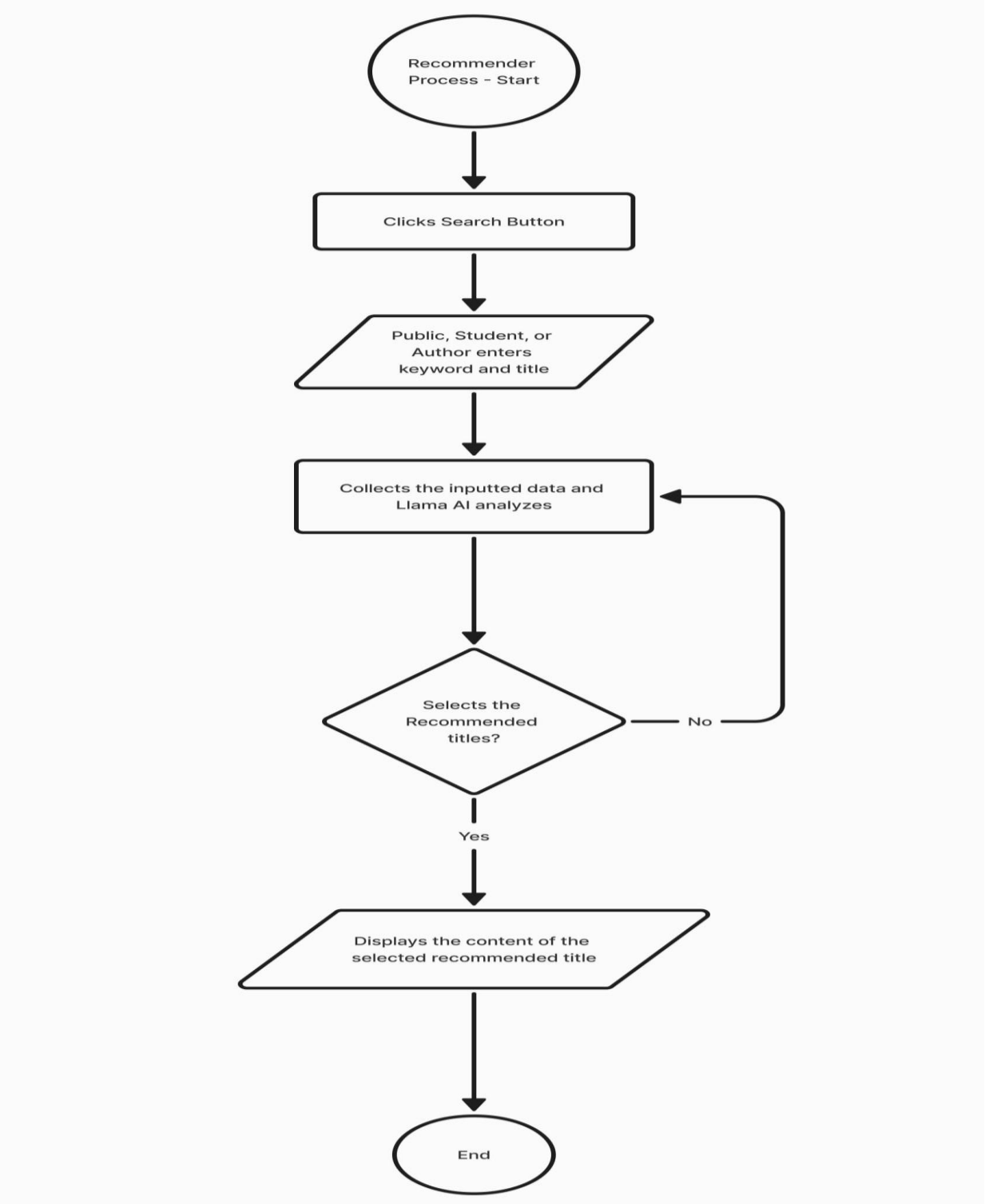

Figure 9. Archiving hub Recommender Process Flowchart

The Recommender System Flowchart begins with user interaction through the Search Button, where the Public, Student, or Author inputs a keyword or title. The system collects this input data and processes it using Llama AI for analysis.

Based on the collected input, the system determines whether the user will Select the Recommended Titles. If the user does not select any title, the process loops back, allowing them to re-enter or refine their search input. If the user selects a recommended title, the system proceeds to display the content of the selected recommended title.

Finally, the process ends, completing the interaction. This structured flow allows for seamless user interaction and efficient retrieval of relevant content while ensuring a smooth and iterative user experience.

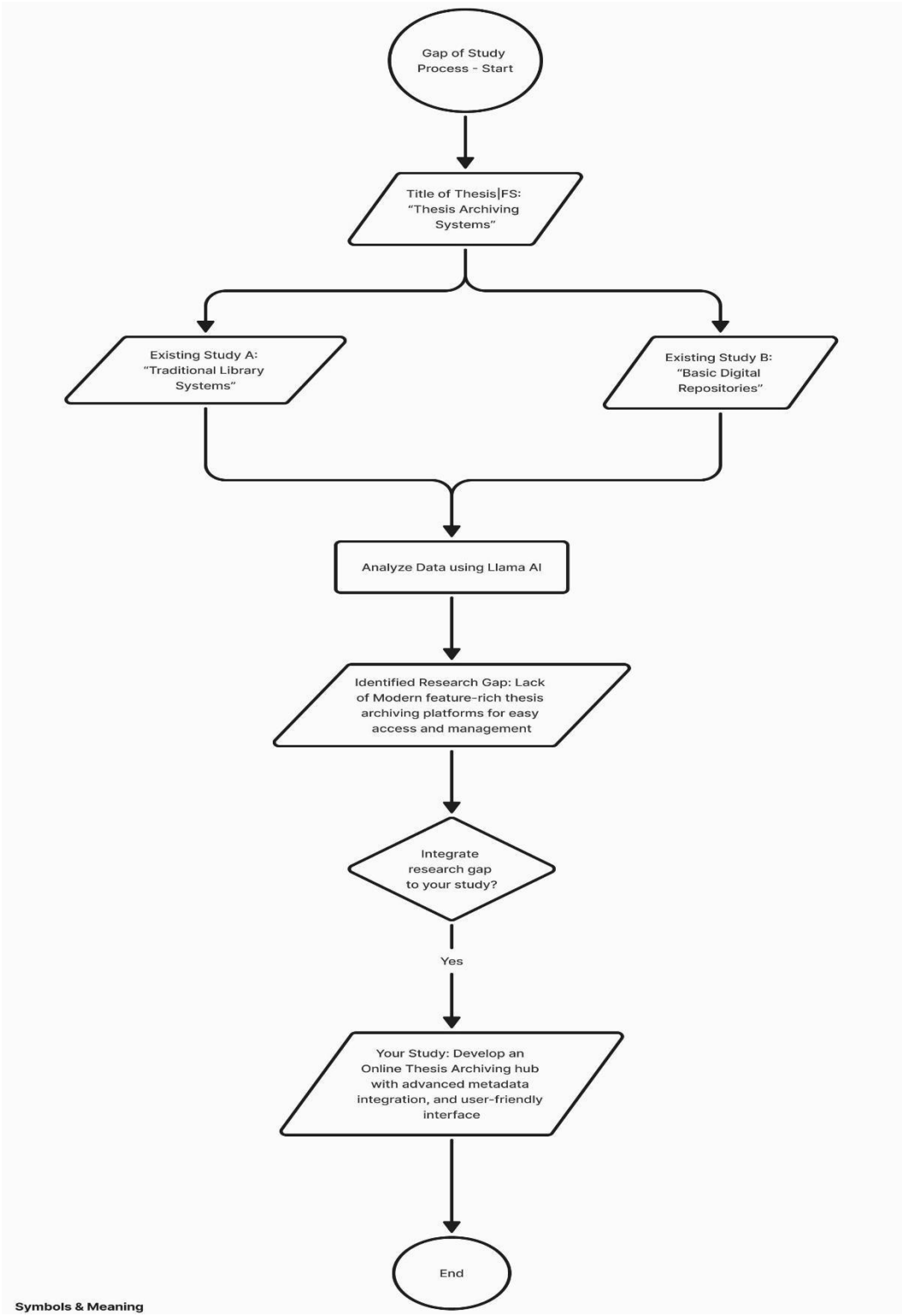

Figure 10. Archiving hub Gap Process Flowchart

Use Case Diagram

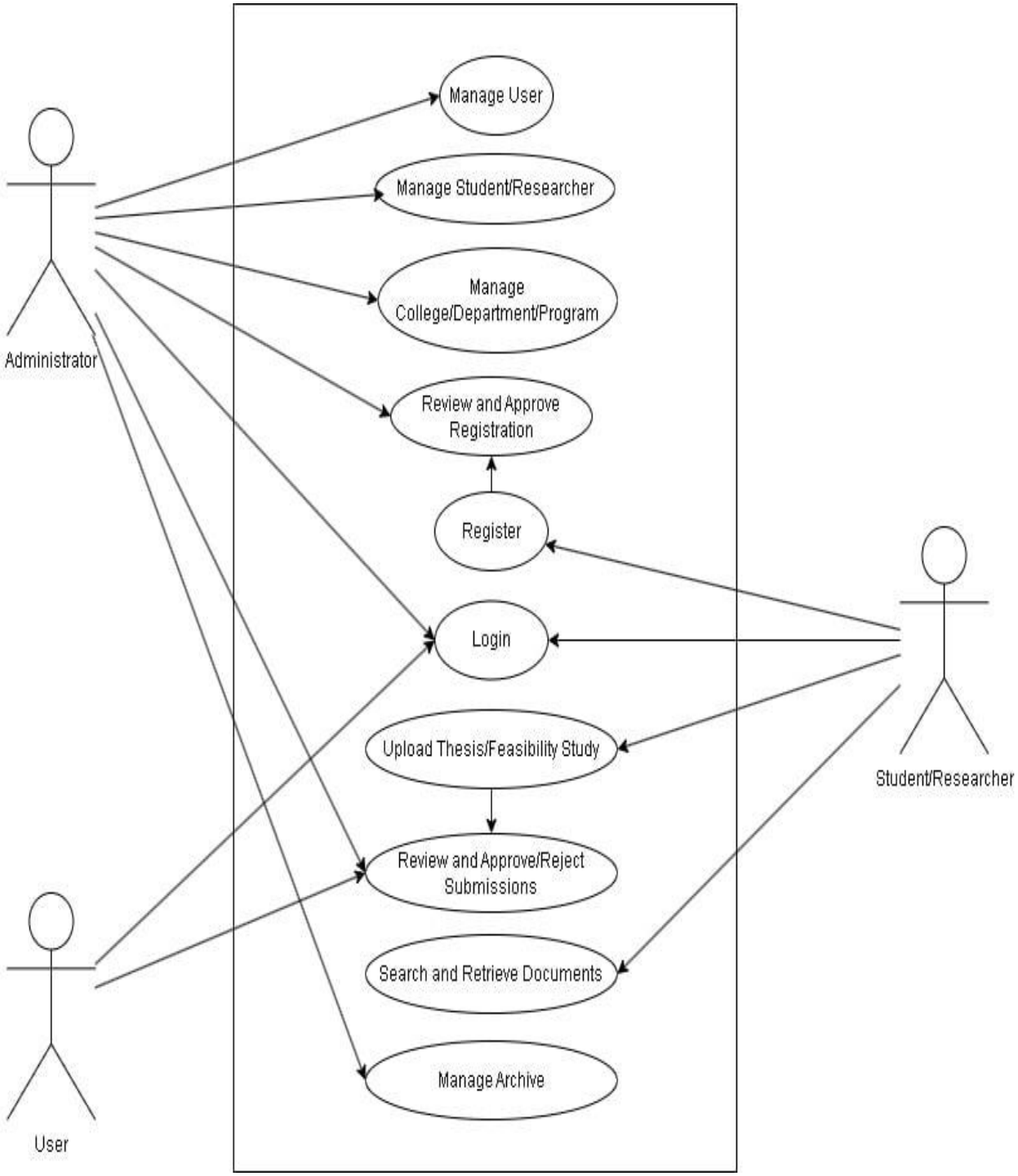

Figure 11. Use Case Diagram

The Use Case Diagram illustrates the interactions between different user roles (Admin, Instructor, Student) and the system. It shows the main functions and functionalities that each user role can perform within the system.

Entity Relationship Diagram (ERD)

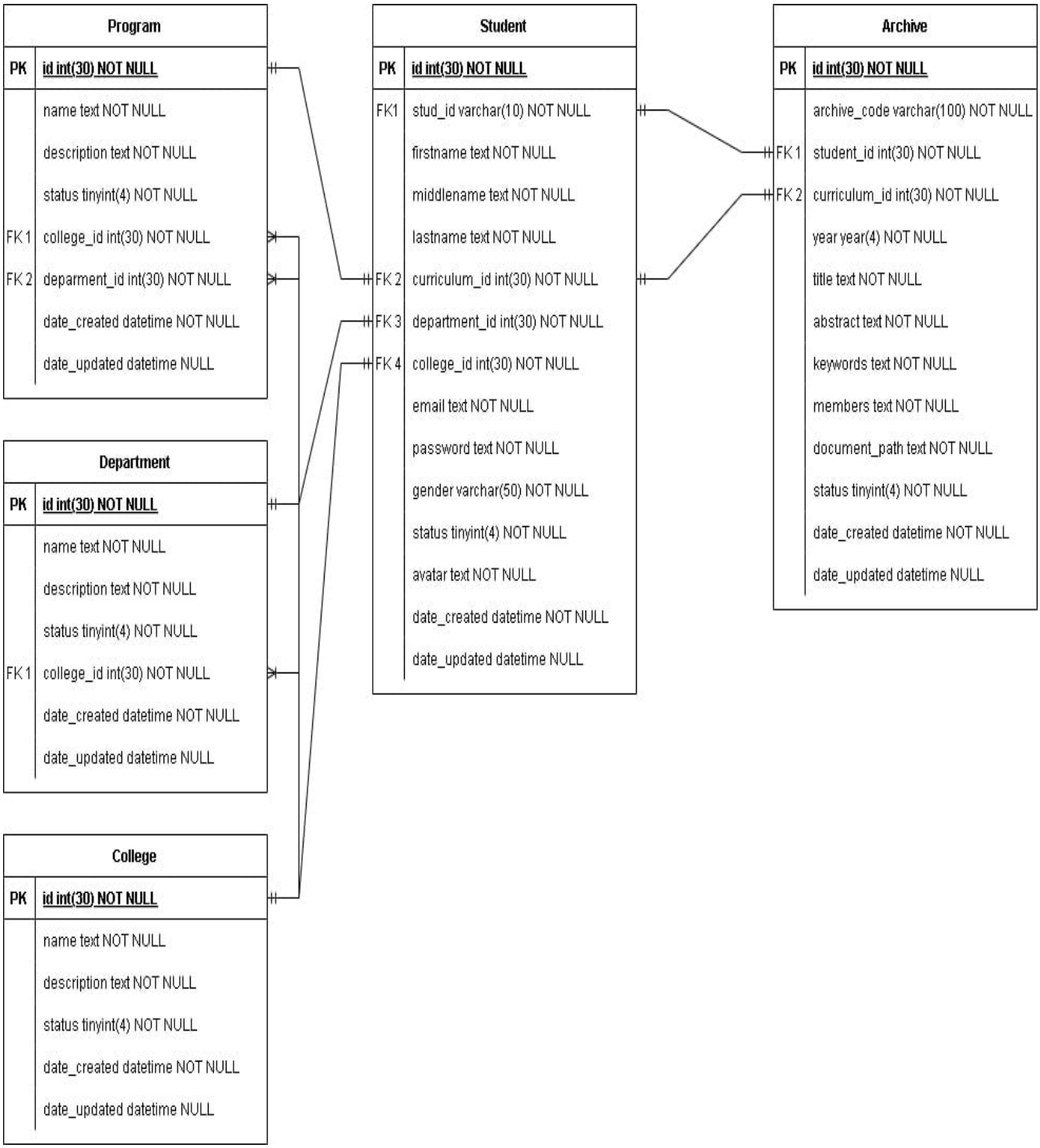

Figure 12. Entity Relationship Diagram (ERD)

The ERD depicts the relationships between different entities in the system, such as students, programs, departments, colleges, and archived research documents. It shows how these entities are connected and how data flows between them.

The Gap Analysis flow focuses on the area of thesis archiving systems. It starts by analyzing existing studies, specifically traditional library systems and basic digital repositories. Through evaluation, the process identifies a significant gap: the lack of modern, feature-rich thesis archiving platforms that offer easy access and efficient management.

This identified gap serves as the foundation for the proposed study, which aims to develop an online thesis archiving hub. The proposed system will feature advanced search capabilities, metadata integration, and a user-friendly interface to improve accessibility and organization of academic theses.

The resulting platform addresses the research gap by enhancing user experience, facilitating streamlined submissions, and ensuring efficient thesis management. This development contributes to the modernization of academic archiving systems, effectively bridging the identified gap.

Phase IV – Evaluate (Review and Development): The evaluation phase is dedicated to gathering feedback and evaluating the functionality of the system. Students, author, and staff test the functionalities of the NEMSU Archiving Hub system in a controlled environment. Before a system is fully implemented, feedback is utilized to determine what needs to be improved to ensure that users' needs are met.

Research Locale

This study was carried out at North Eastern Mindanao State University (NEMSU) Tandag Campus, selected as the main research site for its alignment with the aims of the A Decision Support System Archiving Hub: Transforming Thesis and Feasibility Study Using Llama.The selection of this site was motivated by the institution's vital role as a repository for academic research outputs, particularly feasibility studies and theses, as well as its growing need for new technology to improve archiving and retrieval processes.

NEMSU functions as a crucial environment for evaluating and executing the suggested system, owing to the obstacles given by the conventional manual archiving approach, including inefficiency, risks of data loss, and difficulties in retrieving academic materials. Furthermore, the university offers a rich and pertinent context because of the diverse student body and faculty members who are actively engaged in research.

Participants in this study included academics, students, and university employees in charge of research document management, guaranteeing a thorough comprehension of the needs and functionality of the created archiving hub.

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Figure 13. Location Map of the Study

Research Respondent

The study's research will involve a total of 357 respondents, students from various academic programs at NEMSU Tandag Campus. These respondents will participate in surveys, interviews, and system usability testing. Their feedback and perspectives are essential in evaluating the effectiveness, usability, and practicality of the proposed Decision Support System (DSS) Archiving Hub for academic works.

Table 1. Distribution and Types of Respondents

Respondents	**Population Size**	
	3rd Year	4th Year
CAS Students	62	55
CBM Students	83	71
CITE Students	17	13
CET Students	5	3
CTE Students	28	20
Sub Total	**195**	**162**
Total	**357**	

Data Gathering Procedure

To understand the current manual process of archiving and retrieving academic works at NEMSU Tandag Campus, the research team prepares a formal letter addressed to the university librarian and department heads, requesting permission to conduct the study and gather data. Once approval is granted, the research team conducts interviews with librarians, faculty members, and selected students to assess their experiences, challenges, and recommendations regarding the existing archiving system.

During the interviews, the researchers present the proposed DSS Archiving Hub and seek feedback on its features, usability, and potential areas for improvement. The insights gathered are instrumental in tailoring the system to meet the specific needs of administrators, faculty, and students.

Furthermore, the researchers analyze the current process of archiving thesis and feasibility studies, including submission workflows, categorization methods, and document retrieval practices. These findings provide a comprehensive understanding of the requirements for the proposed system and guide the development of a more efficient and user-friendly digital archiving solution.

Ethical Considerations

Ethical considerations are fundamental in conducting this study, DSS Archiving Hub: Transforming Thesis and Feasibility Study Management. The researchers ensure open and respectful communication with all participants throughout the data gathering process, which involves interviews, surveys, and system usability testing. Full disclosure is provided to all participants, and informed consent is obtained prior to their involvement in the study, in compliance with the Data Privacy Act of 2012 (RA 10173).

The study prioritizes the safety, confidentiality, and well-being of the respondents, ensuring that all personal and academic information is securely stored and used exclusively for the research purpose. Measures are taken to ensure that no data is shared outside the research context, reflecting a strong commitment to ethical practices in academic research.

The researchers adhere to principles of transparency, credibility, and validity during data collection and system development. Collaboration with participants is conducted in a respectful and ethical manner, ensuring that their feedback contributes to creating a user-centered archiving system. The study's implementation also emphasizes fostering a positive relationship with respondents and maintaining trust throughout the research process.

Statistical Treatment

The study employs a quantitative research approach. To analyze the data, purposive sampling and stratified proportional random sampling is utilized to ensure that different subgroups of the population are proportionally represented in the sample. The formula used to determine the sample size for each stratum is:

$$n_i = N_i / N \times n$$ Where:

n_i = the sample size for stratum i

N_i = the population size of stratum i

N = the total population size n =

the total desired sample size

Materials

Table 2. Software Requirements

Name	Purpose
Windows 10/11	Operating System
Browser	To access the Web Based Platform
PHP	Dynamic Content, Interaction with Database, and API development
MySQL	Relational Database Management System
XAMPP	Local Development Environment and Local Web Server
Visual Studio	To access Programming Languages

Table 3. Hardware Requirements

Name	Specification
CPU	2 cores or higher
ROM	128GB SSD or HDD
RAM	8GB
Monitor	Any
Keyboard	Any
Mouse	Any

Schedule and Timeline (Gantt Chart)

The proponents adhered to a schedule and timeline to complete the task, using a work plan and creating a summarized timeline as shown in the table below.

Table 4. Gantt Chart

Activity	**Duration (in Months)** **2024-2025**									
	Aug	**Sept**	**Oct**	**Nov**	**Dec**	**Jan**	**Feb**	**Mar**	**Apr**	**May**
Title formation										
Title Defense										
Formulating the Chapter 1										
System, Prototyping										
Formulating the Chapters 1 & 2										
Pre-Oral Defense										
Data Collection										
System Development										
Crafting Chapters 4 & 5										
Final Oral Defense Preparation										
Plagiarism/Grammarly Checking										
Deployment										

Project Teams and Responsibilities

Every member of the team has responsibility for the system to be successfully created. The Members have their tasks and collaborate in working together on the pieces of the study. All tasks were given individually to achieve the goal of the study.

Proponents	**Responsibility**
Atriz, Wendel Vencent	● Data Provider ● UI Designer ● Note-taker
Dedicatoria, Joven	● Data Provider ● Encoder ● UI Designer
Mendoza, John	● Data Provider ● Note-Taker
Planas, Faustine Frichot G.	● Data Provider ● Encoder ● UI/UX Designer ● Document Organizer
Ponciano, Jackelyn	● Data Provider ● Encoder ● Meeting Minutes Recorder

Budget Cost/Management Plan

The table below shows the budget expenses of this research and system development used as time progresses as well as the cost incurred.

Table 6. Budget Cost Management Plan

Quantity	Unit	Description	Unit Price	Amount
1	App	Laragon	Free	0.00
1	Pdf	Laravel	Free	0.00
2	Sub	Web Hosting	450.00	900.00
1	Unit	Laptop	34,500.00	34,500.00
Total:				**35,400.00**

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